CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 2: **Electricity: Circuits and their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which choice correctly lists materials in order: conductor, insulator, conductor? 1

(a) rubber, iron, plastic **(b)** plastic, copper, wood
(c) iron nail, rubber, aluminium foil **(d)** wood, glass, copper

2. Which effect of electric current is utilised in a fuse? 1

(a) heating effect **(b)** light effect **(c)** sound effect **(d)** chemical effect

3. In an electric kettle, the heating coil is immersed in water yet is insulated by: 1

(a) air
(b) plastic only
(c) nothing
(d) a sealed metal sheath with insulating powder

4. Why is a thick wire used for heavy appliances like geysers? 1

(a) it is cheaper
(b) it does not conduct
(c) it can carry more current without overheating
(d) it is lighter

5. Which of the following is a reason for coiling the heating wire in an electric heater? 1

(a) coiled wire is an insulator **(b)** to cool the wire
(c) a long wire fits in a small space **(d)** to stop current

6. Which of the following may reduce the electricity bill? 1

(a) switching to LED bulbs **(b)** using old heaters
(c) leaving lights on **(d)** using many bulbs

7. Why are electrical sockets kept away from water in kitchens and bathrooms? 1

(a) water makes wires thick **(b)** water is an insulator
(c) water makes sockets cold **(d)** water conducts and can give a shock

8. Which of the following might cause overloading? 1

(a) connecting many appliances to one socket **(b)** switching off
(c) using a fuse **(d)** using one bulb

9. A power cut occurs when a cable gets cut. The circuit becomes: 1

(a) closed **(b)** short **(c)** open **(d)** parallel

10. Which of these is a good habit? 1

(a) touching wires with wet hands
(b) overloading a socket
(c) replacing a blown fuse of the same rating after switching off the supply
(d) using a thick copper wire as a fuse

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a bulb in a circuit glow immediately when the switch is on? 2

12. How are cells connected to increase the voltage? Show with an example. 2

13. Why is it important that wires of the right thickness are used in the house wiring? 2

14. Explain why metals are used for making heating plates in toasters. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how a short circuit can start a fire. 3

16. Why is distilled water not used for the tester but tap water works? 3

17. Describe why wires used for transmission are made of aluminium and not copper in many places. 3

OR

Describe a method to identify the conductors among liquids like lemon juice, sugar solution and salt solution. 3

18. Explain why a current of 4 A makes a wire hotter than 1 A in the same wire. 3

OR

Why must we not use a bulb of higher wattage than the holder's rating? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A bulb rated 4.5 V is to be lit with 1.5 V cells. Describe the circuit, voltage, and what happens if only 2 cells are used. 5

OR

Discuss the advantages and limitations of heating effect of current. 5

20. Discuss the working of a simple tester that can be made with a cell, a bulb and wires, and its use in finding conductors, including limitations. 5

OR

Explain with a circuit diagram description how a fuse is connected and why in the live wire. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND